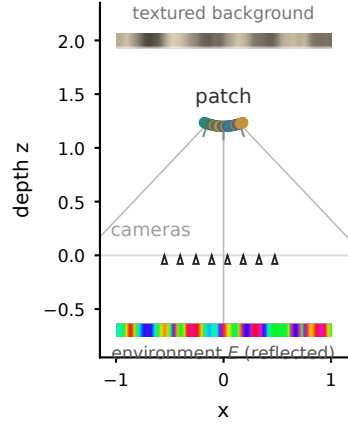
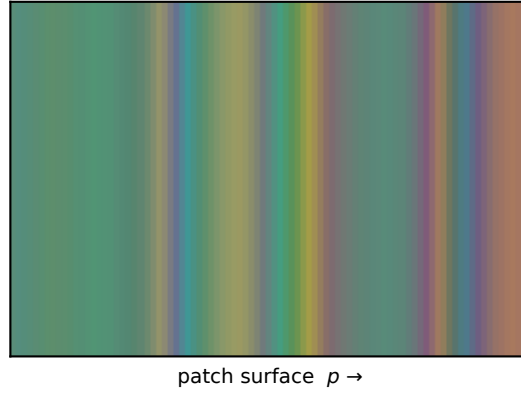


Reflection separation — initial conditions & loss components · $\hat{D} = \text{sigmoid}(\text{logits})$, $\hat{E} = \text{env map}$ · $\alpha = 0.5$, known mask

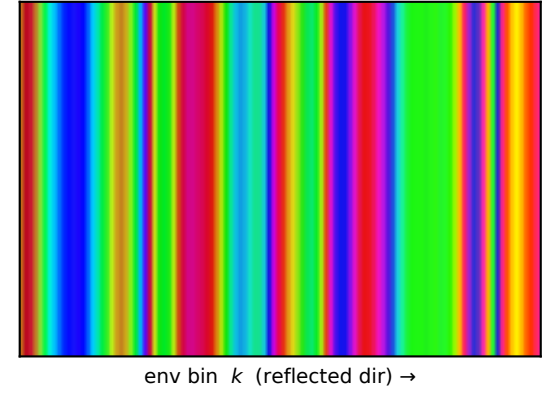
scene: curved glossy patch



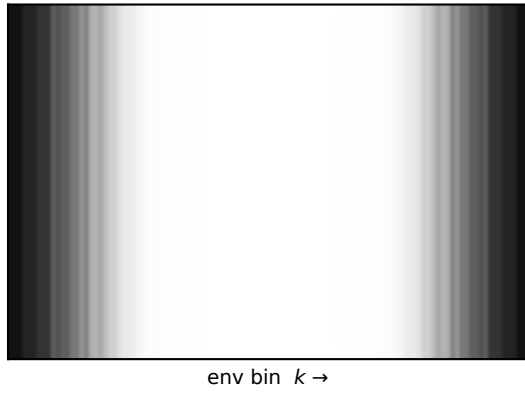
init diffuse $\hat{D}_0 = \text{sigmoid}(\text{logits}_0)$ (per-point mean)



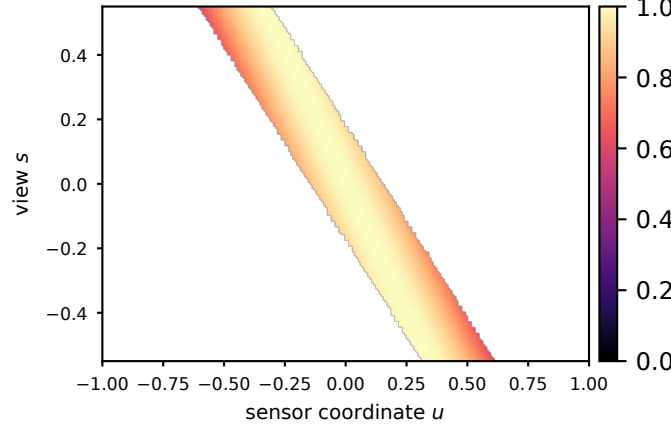
init env $\hat{E}_0$ (splat warm-start of residual reflective est.)



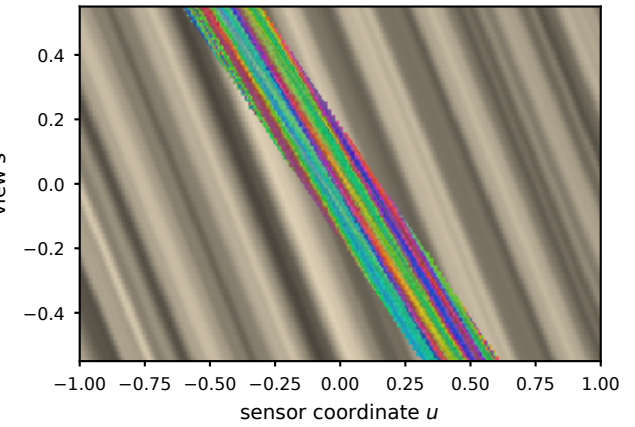
env confidence $c = 1 - e^{-\text{accum}_w/\text{ref}}$ (observed→white)



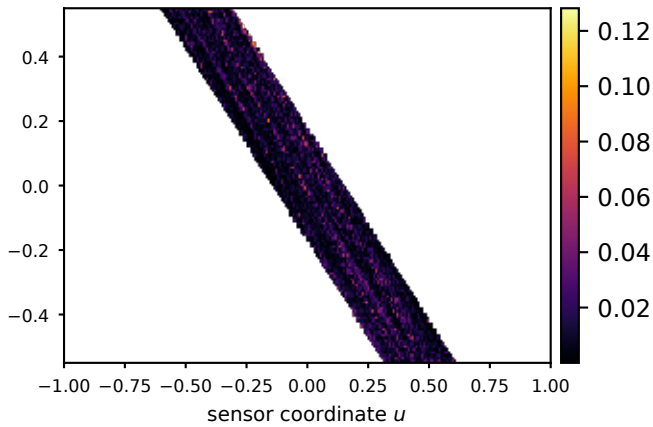
per-obs weight $|\cos(\text{view}, n)|$ (grazing→0)



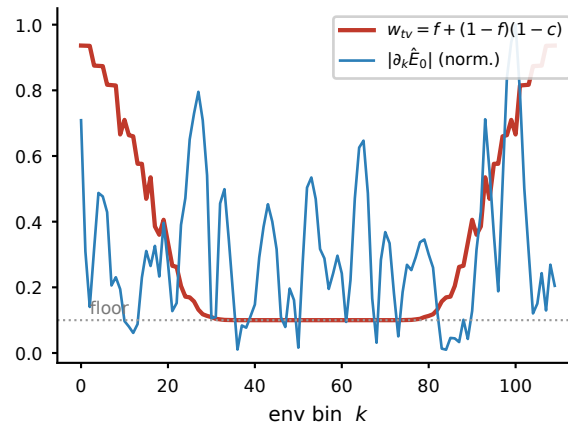
observed input $O = \alpha D + (1 - \alpha)E$ (noisy)



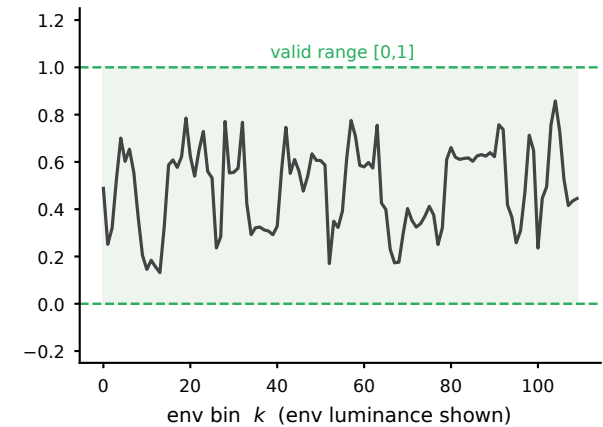
$\mathcal{L}_{\text{recon}}$: weighted residual $w(\hat{O}_0 - O)^2$



$\lambda_{tv} \mathcal{L}_{tv}$: confidence-weighted TV ($\lambda_{tv} = 0.01$)



$\lambda_{rng} \mathcal{L}_{rng} = \lambda_{rng} \text{mean}(\text{relu}(-E) + \text{relu}(E - 1))$ ($\lambda_{rng} = 0.1$)



+ multi-frame prior $\lambda_{pr} \|E - E_{\text{prev}}\|^2$ on observed bins ($\lambda_{pr} = 0.1$; inactive on frame 0)